

The Impact of AI Tools on Academic Productivity Among University Students in Sri Lanka

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Abstract

Artificial Intelligence (AI) technologies have become increasingly popular in the field of higher education due to their ability to support learning activities and improve educational efficiency. University students widely use AI-powered tools such as ChatGPT, Grammarly, and other intelligent learning platforms to assist with learning, information searching, writing improvement, time management, and academic tasks. As the use of AI tools continues to grow among students, it is important to understand how these technologies influence academic productivity in the university environment.

The main objective of this study was to examine the impact of AI tools on academic productivity among university students in Sri Lanka. The study adopted a quantitative research approach, and primary data were collected using an online questionnaire developed through Google Forms. A total of 61 valid responses were collected from students studying in both government and private universities in Sri Lanka. The collected data were analyzed using Microsoft Excel through percentage analysis, mean analysis, charts, and correlation analysis.

The findings of the study revealed that university students frequently use AI tools for academic activities. The overall mean score for AI tool usage was 4.24, indicating strong agreement among respondents. In addition, the overall mean score for academic productivity was 3.99, showing that students generally agreed that AI tools positively contribute to their academic productivity. The Pearson correlation analysis showed a strong positive relationship between AI tool usage and academic productivity, with a correlation value of $r = 0.752$. Therefore, the study confirmed that

increased use of AI tools is associated with higher academic productivity among university students in Sri Lanka.

The study concludes that AI technologies play a significant role in supporting students' learning efficiency, time management, and overall academic performance. However, students should use AI tools responsibly and maintain a balance between AI-assisted learning and independent thinking. The findings of this study may be useful for students, lecturers, universities, and future researchers in understanding the role of AI technologies in higher education.

Keywords: Artificial Intelligence, AI Tools, Academic Productivity, University Students, Higher Education, Sri Lanka

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Chapter 1

Introduction

Artificial Intelligence (AI) is among the most impactful technologies in today's education. Today, students in universities are equipped with AI tools like ChatGPT and Grammarly to enhance their learning experience. Students are getting more and more support from AI tools like ChatGPT, Grammarly, and other smart learning platforms in their university studies. The tools are used to help students grasp the learning materials, enhance the quality of written work, spark ideas for writing, and organize learning tasks efficiently. The advancement of digital technologies has ensured that AI tools are now more available and accepted in the context of Higher Education.

In Universities, students are under much pressure when they are given examinations, presentations, learning activities and/or coursework. As a result, many students use AI to enhance their learning experience and save time. AI tools can be used to instantly respond to inquiries, simplify complex topics, and organize students' academic work more efficiently. Therefore, AI is slowly replacing traditional learning methods with technology, making learning more effective.

Academic productivity is also a significant indicator of academic performance in higher education since it shows the students' ability to finish their academic work efficiently and effectively. By improving productivity, productivity can help with better time management, students' academic performance, and learning outcomes. While there are some benefits to using AI, there are also some concerns, such as depending too heavily on AI and how it affects students' ability to think for themselves. Hence, it is important to study the impact of AI tools on university students' academic productivity in Sri Lanka. (Kasneci, E, et al., 2023) (UNESCO, 2021) (Selwyn, 2019)

Background of the Study

Artificial Intelligence (AI) refers to computer-based technologies that can perform tasks that normally require human intelligence, such as understanding language, generating content, solving problems, and analyzing information. Students at the university have been increasingly using AI tools like ChatGPT, Grammarly, and AI learning systems in recent years. They are used extensively in the education sector due to their ability to access information rapidly, offer writing support, generate ideas, and offer learning support.

The students of the University have different approaches to using AI tools to enhance their learning process. AI applications are utilized by many students for summarizing learning content, creating presentations, enhancing grammar, and time management. AI can also assist students in comprehending complex concepts by offering explanations and tailored learning assistance. With such advantages, the application of AI in the field of higher education has been growing rapidly across the globe, even in Sri Lanka.

The use of AI in universities has brought opportunities and challenges to the educational field. AI tools can enhance efficiency, learning rates, and academic output on the one hand. On the other hand, AI tools can boost efficiency, learning rates, and academic productivity. However, overreliance on AI tools could lead to a decline in critical thinking, creativity, and self-directed learning skills in students. The value of academic productivity lies in its contribution to students' academic achievement, learning outcomes, and learning success. Hence, the true effect of the use of AI tools on academic productivity of university students in Sri Lanka should be studied. (Luckin, R., Holmes, W, Griffiths, M, & Forcier, L. B, 2016) (Zawacki-Richter, O, Marín, V. I, Bond, M., & Gouverneur, F., 2019)

Problem Statement

The field of education has been profoundly transformed by the swift advancement of Artificial Intelligence (AI) technologies in universities across the globe. AI-powered tools such as ChatGPT, Grammarly, and other intelligent learning applications are increasingly used by university students for academic purposes. These tools help students brainstorm, enhance their writing, comprehend complex topics, and save time in their learning process. With the widespread access to the internet and digital technologies, university students in Sri Lanka have also grown fond of AI tools.

While AI tools have numerous benefits, there is still some doubt about their true effect on academics' productivity. AI tools are seen as tools that enhance efficiency, accelerate learning, and enrich learning experiences by some researchers and educators. Some others, however, say that an overreliance on AI tools could have a negative impact on students' critical thinking, creativity, problem-solving and independent learning. This has led to a debate about the actual value of AI on the academic landscape, rather than simply making academics easier.

The second major concern is that lots of university students might be excessively dependent on information generated by AI without actually comprehending the material. This dependence might diminish the actual learning and engagement in school. While the adoption of AI tools in higher education has risen, there is a lack of empirical studies to explore the substantial effects of these tools on academic output among university students in Sri Lanka. There has been less empirical research to study the actual impact of AI tools on university student's academic productivity in Sri Lanka. Thus, it is important to explore the effect of the use of AI tools on academic productivity, efficiency and performance of students in Sri Lankan university environment. To fill this research gap, this study will investigate the relationship between the use of AI tools and academic productivity of the university students in Sri Lanka. (Dwivedi, Y. K., et al., 2023) (Tlili, A, et al., 2023)

Research Objectives

Main Objective

- To examine the impact of AI tools on academic productivity among university students in Sri Lanka.

Sub Objectives

- To identify the level of AI tool usage among university students in Sri Lanka.
- To analyze students' perceptions of the usefulness of AI tools in academic activities.
- To examine how AI tools influence students' learning efficiency and time management.
- To determine the relationship between AI tool usage and academic productivity among university students.

Research Questions

1. What is the level of AI tool usage among university students in Sri Lanka?
2. What are students' perceptions of the usefulness of AI tools in academic activities?
3. How do AI tools influence students' learning efficiency and time management?
4. What is the relationship between AI tool usage and academic productivity among university students in Sri Lanka?

Research Hypotheses

The primary aim of this study is to determine the influence of the utilization of Artificial Intelligence (AI) tools on the academic productivity to the university students in Sri Lanka. The study's independent variable is the use of the AI tools, and the dependent variable is the academic productivity.

Students are increasingly using AI tools like ChatGPT, Grammarly, and other AI learning tools for educational purposes. These tools could be used to enhance the efficiency of a student's learning, save time for study, make difficult concepts more comprehensible, and more effectively organize academic activities. Given these possible benefits, it is crucial to see if there is a significant correlation between the use of AI tools and academic productivity.

The analysis of this research is done under the guidance of two hypotheses. The alternative that the usage of the AI tool has a positive effect on academic productivity is H1, and the null hypothesis is that there is no relationship between the usage of the AI tool and academic productivity. The hypotheses will be tested using the data obtained from the University students by administration of questionnaire and the analysis of the data will be carried out with the help of statistical techniques which are available in MS Excel like correlation analysis.

H₁ (Alternative Hypothesis)

- ❖ There is a positive relationship between AI tool usage and academic productivity among university students in Sri Lanka.

H₀ (Null Hypothesis)

- ❖ There is no relationship between AI tool usage and academic productivity among university students in Sri Lanka.

Significance of the Study

The study is important as it provides a study on the effects of using Artificial Intelligence (AI) tools on the academic productivity of university students in Sri Lanka. As AI technologies continue to develop rapidly, students increasingly use AI-powered tools such as ChatGPT and Grammarly in their educational activities. It is crucial for students, lecturers, universities and future researchers to have an insight into the impact of these technologies on student productivity.

One of the merits of this study is that it will enable university students to understand the pros and cons of using AI tools in academic tasks. The results of the study could inform students on the potential of AI tools in enhancing the efficiency of learning, saving time, and aiding academic achievement. Meanwhile, learners will gain greater understanding of the responsible and non-tech-driven use of AI.

Secondly, this study is important for lecturers and educators. AI technologies have revolutionized the education sector, transforming how teaching and learning are conducted in classrooms. It is essential for lecturers to grasp the usage of AI tools by students and the impact they have on learning activities and productivity. The results of this research can inform educators in creating effective learning approaches and lead learners to engaging and ethical use of AI technologies.

Thirdly, the study is useful for the universities and educational institutions in Sri Lanka. There are several current challenges related to the use of AI in education, such as ensuring academic integrity, integrating technology into the learning process, and developing policies. Insights from this study can inform universities to create appropriate guidelines, policies, and awareness initiatives for the use of AI tools in higher education.

Besides, this study adds to the body of knowledge on the use of AI in education, particularly in Sri Lankan context. This research fills an important gap in the existing literature, as few studies have so far been conducted on the relationship between limited research with Academic productivity of university students in relation to AI tools in Sri Lanka. Lastly, the findings of this study could serve as valuable data for future studies in the field of AI technologies, student learning behavior, and educational productivity. (Popenici, S. A. D & Kerr, S., 2017) (UNESCO, 2021)

Scope of the Study

This study is limited to the effects of the Artificial Intelligence (AI) tools on the academic productivity among university students from Sri Lanka. The study questions specifically include the use of AI-powered tools by university students in their academic pursuits and how those tools affect their learning, time management, and academic productivity.

This is a study of the students of the universities of Sri Lanka only who are studying in the University at the time of the study. Students from various academic disciplines and levels of study may be part of the group of participants. The study however does not concentrate on lecturers, school students or employees. The research is primarily focused on students, as they are one of the most dynamic groups in the educational setting in terms of their use of AI technologies.

The research will be based on popular AI products, including ChatGPT, Grammarly, and other AI-driven learning tools for educational purposes. These tools can be used to help students learn the material, enhance the quality of their writing, brainstorm, and organize study-related tasks. The focus of the research is not on the technical aspects of AI systems, but on the impact of their use on academic productivity.

The study is quantitative research, the method of collecting data is the online questionnaire which will be created using Google Forms. The questionnaire will be distributed among university students in Sri Lanka to gather their opinions and experiences related to AI tool usage. The data that is collected will then be analyzed with the use of statistical techniques in Microsoft Excel that involve percentages, charts, mean analysis and correlation analysis.

Moreover, the analysis in this study only examines the correlation between AI tool usage and academic productivity. In this research, the factors that may affect the students' study performance are not examined in detail, except personal skills, socioeconomic conditions, teaching quality, and psychological factors. Thus, the results of this research are applicable to the sample set and the variables under the research framework.

Limitations of the Study

There are some limitations in the present study which can restrict the overall implications and generalizability of the study. Firstly, the sample size is small which is related to the university students in Sri Lanka. Thus, the results may not be entirely representative of the views and experiences of all university students in the country. Secondly, the study may be limited to students from a few universities or one specific university group and this can minimize the variance of the data collected and the generalizability of the outcome to the entire higher education system in Sri Lanka.

One constraint is that data in the study is self-reported which was obtained from questionnaires, using Google forms. Responses can be inaccurate, incomplete or biased according to the participant's opinions and experiences. Moreover, the study is primarily concerned with the use of AI tools and academic productivity; it is not concerned with other external factors that can have a bearing on students' academic performance.

Chapter 2

Literature Review

Introduction to AI in Education

Artificial Intelligence (AI) is one of the most rapidly evolving technologies of the modern world, with a significant impact on the education sector. AI is the technology and computer systems that are capable of carrying out tasks assigned to them in a manner that humans typically perform using intelligence. Over the past few years, AI technologies have gained traction and become more present in the education field, serving to assist in school teaching, learning, and the management processes.

Advancements in digital technologies and internet accessibility have led to a proliferation of the use of AI in education. AI-related tools are increasingly being used in educational institutions globally to enhance learning experiences and boost educational efficiency. AI tools can give personalized learning aid, instant feedback, automated learning help, and easily accessibility of information. As a result, students are increasingly depending on AI technologies for academic activities.

Students use AI tools like Chat GPT, Grammarly, and AI search engines to ease and facilitate learning activities and make learning more efficient. These technologies aid students in comprehending intricate ideas, enhancing writing skills, structuring study material, and better managing their time. AI can also help students find information easily and get help on the spot, which can enhance the learning experience.

AI is playing a growing role in learning due to its ability to facilitate flexible and technology-driven learning. AI systems have the ability to adjust to the learning needs of students and offer them educational support based on each student's learning style. It has brought a change to the way that things used to be done in learning, and it now involves greater interaction and technology. In higher education, University students are one of the most prolific users of AI technologies, as they are under academic stress regarding learning tasks, exams, presentations, and coursework.

While AI technologies offer numerous benefits, there have been discussions about their overuse in the education sector. There is a concern that over-dependence on AI tools could have a detrimental impact on students' creativity, critical thinking and ability to learn independently. In the contemporary educational landscape, it is vital to understand how AI tools influence students' academic productivity, making it a significant research focus. (Kasneci, E, et al., 2023) (UNESCO, 2021) (Selwyn, 2019)

AI Tools Used by Students

Many of the AI tools available for university use today facilitate university activities. These tools provide various functions such as content generation, grammar correction, idea development, information searching, and learning assistance. AI technologies are becoming increasingly popular, and as such, these tools are very accessible to students globally, including Sri Lanka.

ChatGPT is one of the most popular AI tools among students. ChatGPT is an artificial intelligence (AI) chatbot that can respond to questions and give explanations, summaries and suggestions that resemble humans. ChatGPT serves as a valuable resource for many students, helping them grasp complex topics, thinking through ideas, summarizing study content, and getting immediate explanations and assistance. The ability of ChatGPT to provide quick and interactive responses has made it highly popular in educational settings. This is used by students frequently as it saves time and helps to make difficult academic content easier to understand.

Another commonly used AI tool is Grammarly. Grammarly is an AI-powered writing tool designed to help students enhance their grammar, spelling, sentence construction, clarity, and writing style. Grammarly is utilized by lots of university students in preparing academic documents, presentations, and written assignments. The tool automatically suggests and corrects, helping students to improve their writing skills and minimize the errors associated with their language. With English being a common language in the higher education sector, Grammarly has proven to be a valuable tool for students seeking to enhance their academic communication.

Students utilize AI-powered tools for searching and accessing information, such as ChatGPT and Grammarly, and intelligent learning platforms. AI search capabilities enable students to locate

educational resources that are relevant, summarize text, and better structure learning content. Some AI-powered educational platforms also provide personalized learning experiences based on students' progress and learning behavior. These technologies facilitate efficient learning by assisting students to focus on the most meaningful information and decreasing search and study time.

The adoption of AI tools by students shows how digital technologies are changing the landscape of education. The use of AI in education is becoming more commonplace in students' daily learning experiences, as it offers convenience, access, and learning support. But, as technology becomes more prevalent, concerns have also emerged about academic integrity, plagiarism, and the diminishment of students' independent thinking abilities. Hence, researchers are still studying positive and negative impacts of the use of AI tools in higher education. (AlAfnan, M. A, Dishari, S, Jovic, M, & Lomidze, K., 2023) (Kasneci, E, et al., 2023) (Tlili, A, et al., 2023)

Academic Productivity

Academic productivity is the capability of the student to be able to undertake the educational activities in an effective and efficient manner within a certain time span. Academic productivity is also crucial in higher education as it has a direct impact on students' academic performance, learning outcomes, and educational success. Students who are productive tend to be more effective in managing their time, comprehending what they are learning, doing schoolwork well and attaining educational outcomes well.

Learning efficiency, time management, academic engagement, and educational performance are some commonly used terms that are associated with academic productivity. Learning efficiency is the ability of students to understand and memorize the education that is less demanding and less time. Time management is also a significant factor in student productivity, as students may have different academic duties, such as examinations, presentations, academics, and independent study activities. Time management can help students manage their academic activities and lessen academic stress.

The use of AI technologies could potentially drive academic productivity through efficiency in completing learning activities. AI tools can help make complex concepts easy to understand, offer instant educational support and save time in accessing information. The use of AI technologies might help students become more efficient at learning, as they will receive fast explanations and structured information. For instance, AI-driven learning systems can aid students in comprehending challenging topics more effectively and enhance self-study capabilities.

Another important one of these factors is time saving, related to academic productivity. With the hectic schedules and multiple responsibilities of the university students, they often have less time to do their academic work. AI-powered tools can streamline learning for students by giving them instant answers, editing their writing, and structuring educational material. This can thereby increase student productivity and efficiency in learning.

But being productive is not just about getting things done in an expedient manner. But meaningful learning, gaining knowledge of educational concepts, and critical thinking are also part of true productivity. Overreliance on AI tools could impact students' independent learning and problem-solving abilities, some researchers say. For this reason, it is critical to consider whether AI technologies truly enhance academic productivity or make academic activities easier to execute.

In general, academic productivity is still a significant issue in educational research as it contributes to the success of students, the quality of their learning and their career development. With the continued development of AI technology in higher education, it has become critical to grasp the relationship between AI and academic productivity. (Popenici, S. A. D & Kerr, S., 2017) (Zawacki-Richter, O, Marín, V. I, Bond, M., & Gouverneur, F., 2019) (UNESCO, 2021)

Relationship Between AI and Learning

Artificial Intelligence (AI) has emerged as a hot topic in contemporary learning. The incorporation of AI technologies into education is growing as it offers opportunities to enhance learning experiences, boost efficiency and help students' learning activities. AI has the potential to have a positive impact on learning by delivering personalized educational support, real-time feedback, and convenient access to information.

Among its primary positive impacts on learning, AI can enhance the efficiency of learning. With the help of AI tools, explanations, summaries, and educational guidance can be provided in a short time, enabling students to understand learning materials more effectively. If students experience problems in studying, they can get immediate aid which in turn may help build confidence and minimize the learning obstacles. AI systems can also get customized to students' studying requirements and provide help based on the students' studying styles.

One other beneficial impact is time saving. Students can use AI-generated tools to structure information, enhance their writing, and make complex concepts easier to understand. This helps students to learn more efficiently and more time to dedicate to learning important topics. AI helps in flexible learning as learners can get educational help anytime and anywhere thanks to digital devices and internet.

Further, AI technologies can enhance student engagement and motivation. Interactive AI systems provide more dynamic learning and motivate students to be active in learning processes. AI-driven learning systems can additionally supply tailored recommendations and monitor advancement, possibly assisting students in achieving educational success. AI systems for education can also offer individualized suggestions and monitor learning progress, potentially helping students achieve educational success.

Though there are advantages, there are some potential risks of using AI in education identified. There is a significant worry around reliance on AI technologies. Students could overlook the need to apply their own critical thinking and problem-solving skills, choosing instead to heavily depend on AI tools. Overreliance on AI systems could lead to a decrease in creativity, self-education and analytical abilities of students.

Plagiarism and academic integrity are another significant issue. Some students may be tempted to submit AI-generated content without understanding or originality. There is a possibility that when students are encouraged to create content quickly using AI tools, students will submit AI-generated content without understanding or originality. This lends moral issues in the education system and diminishes the true learning experiences. As a result, universities and teachers are becoming more and more worried about ensuring academic integrity, while keeping students up to date with the modern world.

Moreover, the information generated by AI may also not always be accurate and reliable. AI systems can sometimes give false or misleading information, negatively influencing students' learning of educational content. Students may misunderstand or learn incorrect information if they trust the answers, they receive from AI without verifying them.

In general, the synergy between AI and learning is a mixture of opportunities and challenges. AI tools offer benefits in enhancing learning efficiency and academic productivity but also come with potential risks of dependency, plagiarism, and diminished independent study. In higher education settings, it is crucial to use AI tools responsibly and appropriately. (Dwivedi, Y. K., et al., 2023) (Tlili, A, et al., 2023) (Kasneci, E, et al., 2023) (Selwyn, 2019)

Research Gap

While there are numerous studies conducted internationally to analyze the implications of AI (artificial intelligence) technologies in the field of education, there is still a lack of research especially that examines how the use of AI technologies relates with academic productivity among university students in Sri Lanka. The majority of current research focuses on the general adoption of technology, online learning systems, or the general effect that digital technologies have on education. There are fewer studies that examine directly how AI-based tools affect students' academic productivity, learning efficiency, and educational performance in the higher education setting in Sri Lanka.

In recent years, AI tools such as ChatGPT and Grammarly have become increasingly popular among university students worldwide. But most previous studies have been conducted in developed countries which have different technological infrastructure, educational levels and levels of digital literacy as compared to Sri Lanka. Therefore, results of foreign studies might not be entirely applicable to Sri Lankan university students.

Moreover, the majority of existing studies have focused on the technical abilities and educational value of AI technologies but not on the real effect of such tools on students' academic productivity. There is less focus on learning efficiency, and understanding whether AI tools have a genuinely

positive impact on learning and education, or simply add to the way learners and educators rely on technology.

One of the other areas of research that need to be explored is the ethical and behavioral implications of the use of AI by students. Although a few studies have touched on the issues of plagiarism and excessive reliance on AI systems, there is not much research conducted to examine the perceptions and usage of AI technologies among Sri Lankan university students in their academic pursuits.

In view of this scenario, the purpose of this study is to address this gap of research and investigate the effect of AI tools on academic productivity among university students in Sri Lanka. The study fills the existing void by offering empirical evidence in the Sri Lankan university setting and aiding researchers, educators, and educational institutions improve their understanding of the role of AI technologies in higher education. (Zawacki-Richter, O, Marín, V. I, Bond, M., & Gouverneur, F., 2019) (Dwivedi, Y. K., et al., 2023)

Chapter 3

Research Methodology

Research Design

This research looks at how Artificial Intelligence tools affect university students' academic work in Sri Lanka, using a quantitative approach. We chose this method because it lets us collect numbers and analyze how different things are related, which is perfect for understanding how AI tools impact academic productivity. By using statistics, we can find patterns and trends that show how AI tools are connected to how well students do in school.

This research is looking about and use AI technology at how students feel It uses surveys. with set questions to collect approach is helpful data. The quantitative because it lets the researcher analyze the data quickly using here percentages, averages statistics like, and charts. This approach is also for looking at how AI tools affect academic good which is the main goal of the study. By productivity, and statistics, the researcher can see using numbers different things how understand the data are related and better. This makes it a for achieving what good choice set out to do. the research

This study uses a two-part approach, methods combining descriptive and correlational research. The descriptive part focuses students use AI and on understanding how what they think about it, while the for a connection correlational part looks between using AI tools and how well students do in their studies. By doing this, the research aims to get a clear picture of how AI affects students' academic performance. University. (Creswell, 2014) (Saunders, M., Lewis, P, & Thornhill, A, 2019)

Data Collection Method

For this study, data gathering is done using primary data collection techniques, which involve the collection of data from the target population. Primary data is defined as first-hand information collected for the purposes of conducting research. In this case, the researcher collects data using a questionnaire, which is developed using Google Forms.

Questionnaire method is chosen because it is among the most efficient ways of collecting data from many respondents at once. Questionnaires can easily be sent to numerous participants through emails, social networks, or any other online platform without involving huge costs.

The questionnaire includes several close-ended questions that are developed using the Likert Scale approach. The respondents are asked to state their level of agreement or disagreement on specific statements about the use of AI tools in enhancing academic productivity. Likert scale measures respondents' views, opinions, and experiences in a structured way. Sections of the questionnaire include demographic data, use of AI tools, learning effectiveness, time management, and academic productivity.

In this case, the Google form questionnaire will be circulated through various online platforms, including email, WhatsApp groups, and social networking sites. (Saunders, M., Lewis, P, & Thornhill, A, 2019) (Sekaran, U & Bougie, R, 2016)

Population

In this case, the population under study is university students in Sri Lanka. In other words, this will be composed of undergraduate students enrolled in both private and public universities in various disciplines. University students have been considered as the population for this study since they are one of the biggest users of AI technology within the educational sphere.

AI technology has become a part of student life. Many university students use such AI technologies like ChatGPT, Grammarly, etc., to achieve their academic goals. In this context, it can be argued that students are the perfect source of information and experience in relation to this study.

It should be noted that school students, lecturers, and university administrators have not been chosen to conduct this study as it focuses on examining the effects of AI technology on students' academic productivity. Age groups, years of study, and discipline are not important aspects at this point. (Creswell, 2014)

Sample Size

Sample sizes for this study will lie in the range of 50 to 100. This selected sample size will be appropriate for this study on quantitative research conducted by undergraduates because this sample size gives room for the researcher to collect enough data for statistical analysis, without creating unnecessary data collection and analysis processes.

This kind of sample size will provide sufficient data for conducting descriptive statistical analysis, percentage analysis, mean analysis, and correlation analysis in Microsoft Excel. Furthermore, collecting feedback from more than one university student will ensure more reliability and validity of the research results.

I plan to get responses from students studying in various universities in Sri Lanka. More sample sizes could have given reliable findings but, the above-mentioned range will be adequate enough for this study due to time constraints. (Sekaran, U & Bougie, R, 2016)

Sampling Method

The present research employs convenience sampling as the sampling strategy. Convenience sampling is a type of non-probability sampling wherein the participants are chosen based on their availability and readiness to take part in the study. Convenience sampling is widely used in undergraduate research due to its simplicity and affordability.

I will choose university students who can be contacted personally, via university clubs, social networking sites, and online communication channels. As university students frequently utilize online channels, convenience sampling will allow me to gather information from participants conveniently through online surveys.

Convenience sampling is deemed suitable for this study, as the research primarily aims at collecting general perceptions and experiences concerning the use of AI tools by university students in Sri Lanka. While convenience sampling cannot accurately represent all university

students, it helps me collect adequate information within the given time and resources. (Saunders, M., Lewis, P, & Thornhill, A, 2019)

Data Analysis Tool

The data collected will be analyzed by making use of the Microsoft Excel. Microsoft Excel will be the primary source of analysis due to its wide application and usability in quantitative data analysis. Microsoft Excel allows the creation of various tools for data organization, calculations, statistical data analysis, and graphic display.

In the course of analyzing the data, I will input the information collected from the questionnaires into Excel sheets where calculations such as calculating percentages, means, frequencies, and correlation will be done. Apart from that, Microsoft Excel can allow one to make charts such as bar graphs and pie charts that will aid in graphically displaying the findings of the study in a comprehensible way.

Microsoft Excel will be ideal for the analysis in this study since the sample used in the research will be moderate in size. (Creswell, 2014)

Data Analysis Techniques

This study will use several data analysis techniques to analyze the collected information effectively. These techniques enable me to interpret the data and find associations between the use of AI tools and academic productivity.

Firstly, demographic analysis and students' responses to the usage of AI tools will be analyzed through percentage analysis. Percentage analysis is a method of summarizing and presenting data in a simple and understandable form.

Secondly, mean analysis will be used to calculate the average responses of students related to AI tool usage, learning efficiency, time management and academic productivity. The average values are useful for recognizing the general opinions and attitudes of the respondents.

Thirdly, graphical presentation methods like bar charts and pie charts will be used to present the data collected visually. Charts make research findings more attractive and easier to interpret and help in better understanding of data patterns.

Finally, correlation analysis will be employed to investigate the relationship between AI tool usage and academic productivity. Correlation analysis is used to determine whether there is a positive or negative relationship between the independent and dependent variables. This analysis is important because it provides direct support to the research hypothesis of the study. (Sekaran, U & Bougie, R, 2016) (Creswell, 2014)

Conceptual Framework

The conceptual framework of this study shows the relationship between the independent variable and the dependent variable. The framework helps to provide a clear understanding of the structure of the research and guide the overall study.

Independent Variable (IV)

- AI Tool Use

Dependent Variable (DV)

- Academic Productivity

The independent variable in this study is the use of the AI tool as it is the factor that is expected to influence students' academic productivity. The academic productivity is the dependent variable, which may vary with the students' use of AI technologies.

The conceptual framework posits that the growing adoption of AI tools could impact students' learning efficiency, time management and overall academic productivity. Therefore, the study aims to investigate the relationship between these two variables among university students in Sri Lanka. (Popenici, S. A. D & Kerr, S., 2017) (Kasneci, E, et al., 2023)

Chapter 4

Data Analysis and Results

Introduction

This chapter presents the analysis and interpretation of data collected from university students in Sri Lanka regarding the impact of AI tools on academic productivity. The data was collected through a questionnaire and analyzed using Microsoft Excel. The chapter includes demographic analysis, AI tool usage analysis, academic productivity analysis, and correlation analysis between AI tool usage and academic productivity. Frequency tables, percentages, mean values, and correlation results are used to present and interpret the findings clearly.

A total of 61 valid responses were analyzed in this chapter.

Demographic Analysis

This section presents the demographic information of the respondents, including gender and university type.

➤ Gender Distribution

The following table presents the gender distribution of respondents who participated in the study.

Table 1/Gendar Distribution

Gender	Frequency	Percentage
Male	40	65.57%
Female	19	31.15%
Prefer not to say	2	3.28%
Total	61	100%

Interpretation

According to the findings, 65.57% of the respondents were male students, while 31.15% were female students. A small percentage of respondents, 3.28%, preferred not to state their gender. This shows that the majority of participants in the study were male students.

Chart Analysis

The pie chart illustrates the gender distribution of the respondents who participated in the study. According to the results, out of 61 respondents, 40 were male students, representing 65.57% of the sample, while 19 were female students, representing 31.15%. In addition, 2 respondents, representing 3.28%, preferred not to state their gender. These findings show that the majority of participants were male students. Therefore, the sample can be considered male-dominant, although female students also made a meaningful contribution to the study.

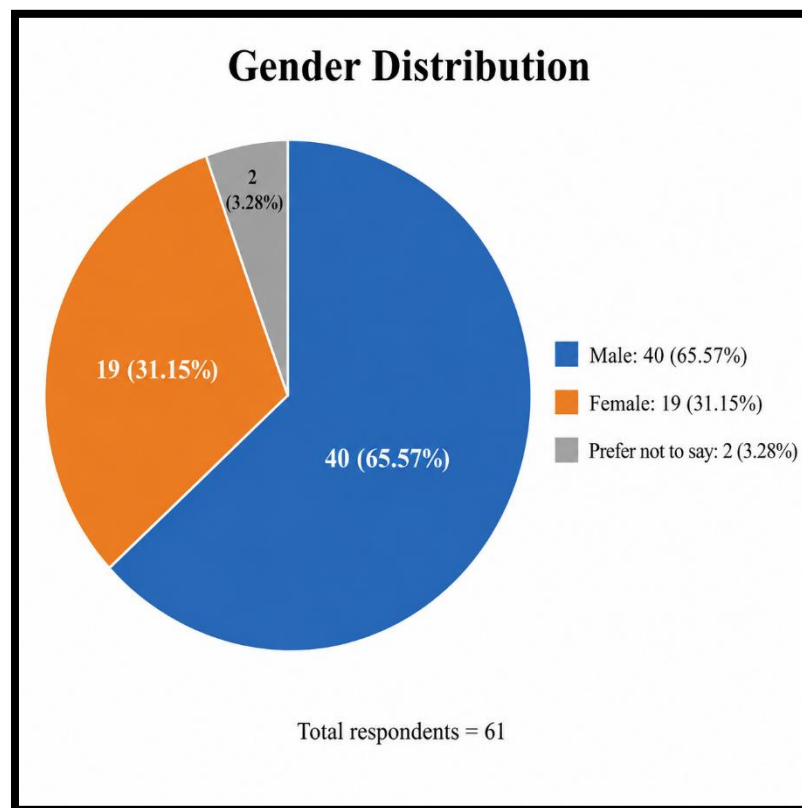


Figure 1/Pie chart for Gender Distribution

➤ University Type Distribution

The following table shows the university type of respondents

Table 2/University Type Distribution

University Type	Frequency	Percentage
Government University	30	49.18%
Private University	31	50.82%
Total	61	100%

Interpretation

The findings indicate that 49.18% of the respondents were from government universities, while 50.82% were from private universities. This shows that the sample included students from both government and private universities almost equally, with a slightly higher number of respondents from private universities.

Chart Analysis

The pie chart shows the distribution of respondents according to university type. Among the 61 respondents, 30 students were from government universities, representing 49.18%, while 31 students were from private universities, representing 50.82%. The results indicate that the sample was almost equally distributed between government and private university students. Private university respondents were slightly higher than government university respondents, but the difference was very small. This balanced distribution is important because it allows the study to include the views of students from both major types of universities in Sri Lanka.

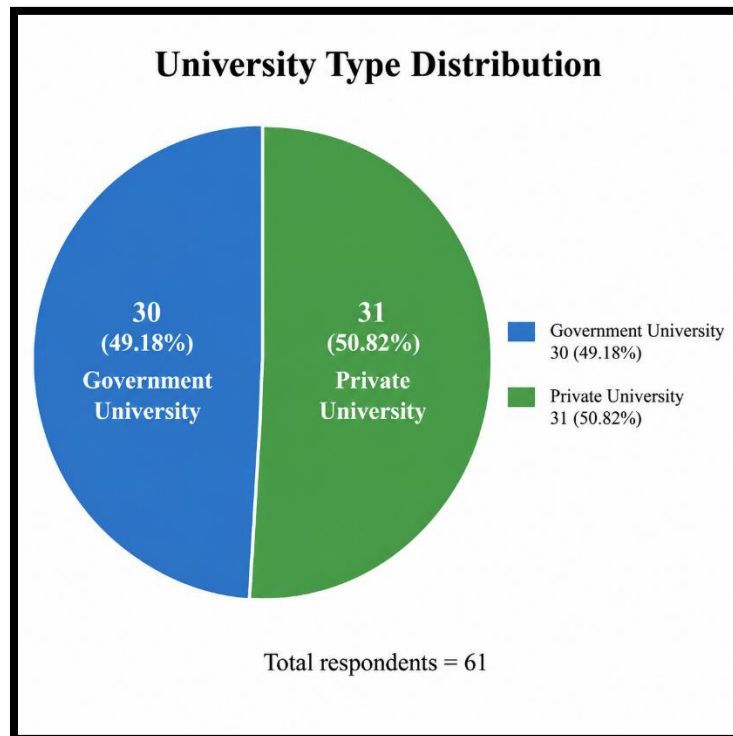


Figure 2/Pie chart for University Type Distribution

AI Tool Usage Analysis

This section analyzes students' responses related to AI tool usage in academic activities. AI tool usage was measured using questions Q3 to Q7.

➤ Analysis of AI Tool Usage Frequency

Question 3: "I frequently use AI tools for academic activities."

Table 3/Analysis of AI Tool Usage Frequency

Response Value	Meaning	Frequency	Percentage
5	Strongly Agree	27	44.26%
4	Agree	26	42.62%
3	Neutral	7	11.48%
2	Disagree	1	1.64%
1	Strongly Disagree	0	0.00%

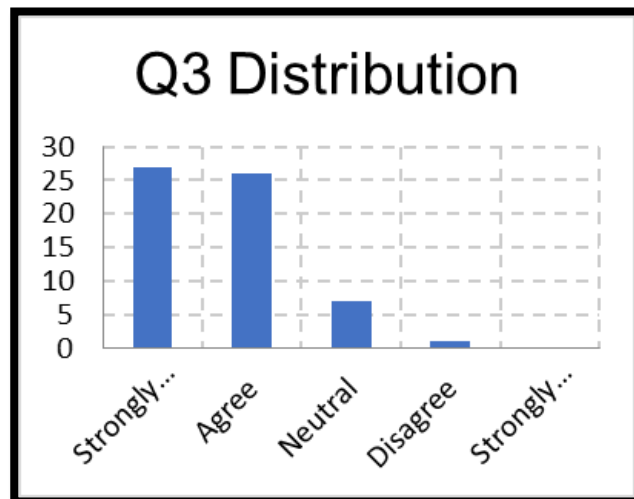


Figure 3/Bar graph for Analysis of AI Tool Usage Frequency

Mean Value = 4.30

Interpretation

The mean value for this question was 4.30, indicating strong agreement among respondents. This shows that most university students frequently use AI tools for academic activities. Therefore, AI tools appear to be commonly used by students in their academic work.

➤ Analysis of Learning Support Through AI Tools

Question 4: “AI tools help me understand learning materials more effectively.”

Table 4/Analysis of Learning Support Through AI Tools

Response Value	Meaning	Frequency	Percentage
5	Strongly Agree	25	40.98%
4	Agree	31	50.82%
3	Neutral	5	8.20%
2	Disagree	0	0.00%
1	Strongly Disagree	0	0.00%

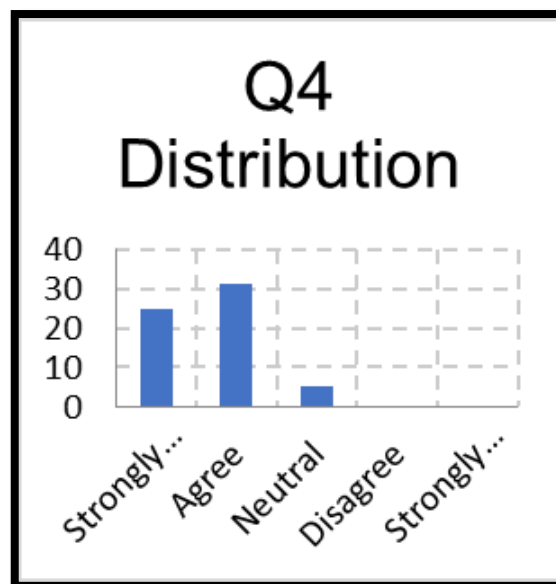


Figure 4/Bar graph for Analysis of Learning Support Through AI Tools

Mean Value = 4.33

Interpretation

The mean value of 4.33 indicates strong agreement. The majority of respondents agreed that AI tools help them understand learning materials more effectively. This suggests that students perceive AI tools as useful learning support systems.

➤ Analysis of Learning Efficiency

Question 5: “I use AI tools to improve my learning efficiency.”

Table 5/Analysis of Learning Efficiency

Response Value	Meaning	Frequency	Percentage
5	Strongly Agree	30	49.18%
4	Agree	24	39.34%
3	Neutral	6	9.84%
2	Disagree	1	1.64%
1	Strongly Disagree	0	0.00%

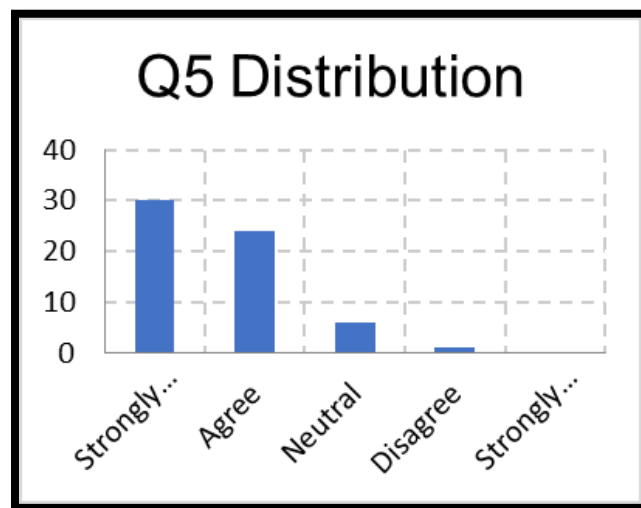


Figure 5/Bar graph for Analysis of Learning Efficiency

Mean Value = 4.36

Interpretation

The mean value for this question was 4.36, showing strong agreement. This indicates that most respondents believe AI tools help improve their learning efficiency. Therefore, AI tools may support students in learning faster and more effectively.

➤ Analysis of Time Management

Question 6: “AI tools help me manage my study time effectively.”

Table 6/Analysis of Time Management

Response Value	Meaning	Frequency	Percentage
5	Strongly Agree	21	34.43%
4	Agree	28	45.90%
3	Neutral	9	14.75%
2	Disagree	3	4.92%
1	Strongly Disagree	0	0.00%

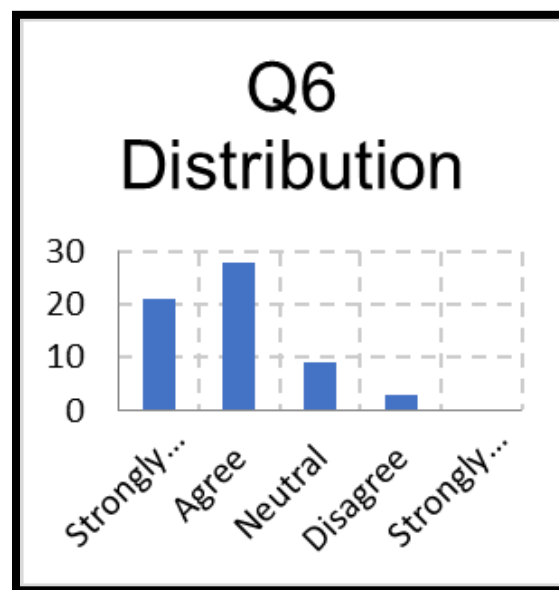


Figure 6/Bar graph for Analysis of Time Management

Mean Value = 4.10

Interpretation

The mean value of 4.10 indicates strong agreement. This shows that students generally believe AI tools help them manage their study time effectively. AI tools may assist students in organizing academic tasks and completing work more efficiently.

➤ Analysis of Task Completion Speed

Question 7: “AI tools help me complete academic tasks more quickly.”

Table 7/Analysis of Task Completion Speed

Response Value	Meaning	Frequency	Percentage
5	Strongly Agree	21	34.43%
4	Agree	26	42.62%
3	Neutral	13	21.31%
2	Disagree	1	1.64%
1	Strongly Disagree	0	0.00%

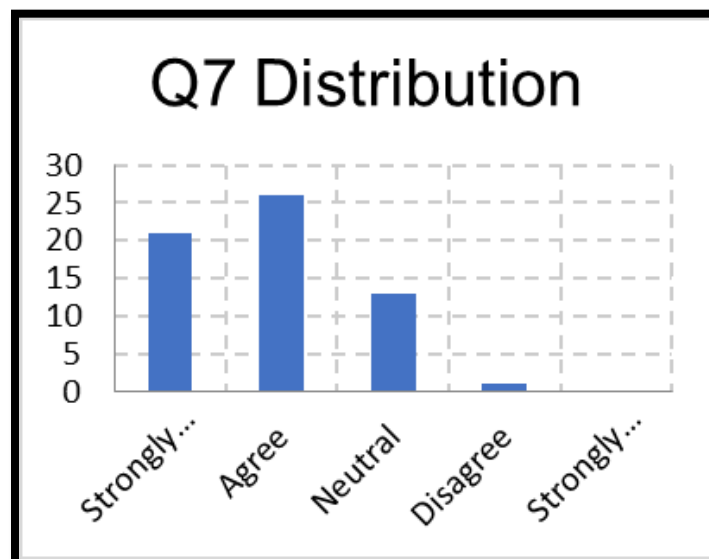


Figure 7/Bar graph for Analysis of Task Completion Speed

Mean Value = 4.10

Interpretation

The mean value for this question was 4.10, indicating strong agreement. The findings suggest that most students believe AI tools help them complete academic tasks more quickly. This highlights the role of AI tools in supporting faster academic task completion.

Overall, AI Tool Usage Mean

Table 8/Overall, AI Tool Usage Mean

Variable	Questions	Mean	Interpretation
AI Tool Usage	Q3–Q7	4.24	Strong Agreement

The overall mean score for AI tool usage was 4.24, which indicates strong agreement. This shows that respondents generally use AI tools actively and positively in their academic activities.

Academic Productivity Analysis

This section analyzes the impact of AI tools on students' academic productivity. Academic productivity was measured using questions Q8 to Q12.

➤ Academic Productivity Improvement

Question 8: "Using AI tools has improved my academic productivity."

Table 9/Academic Productivity Improvement

Response Value	Meaning	Frequency	Percentage
5	Strongly Agree	20	32.79%
4	Agree	30	49.18%
3	Neutral	10	16.39%
2	Disagree	1	1.64%
1	Strongly Disagree	0	0.00%

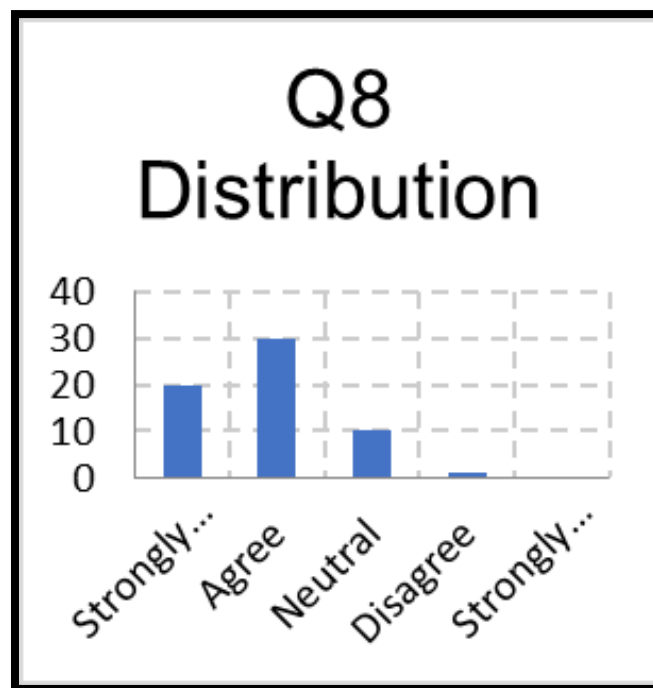


Figure 8/Bar graph for Academic Productivity Improvement

Mean Value = 4.13

Interpretation

The mean value of 4.13 indicates strong agreement. The majority of respondents agreed that using AI tools has improved their academic productivity. This suggests that AI tools have a positive influence on students' ability to complete academic work productively.

➤ Quality Improvement

Question 9: "AI tools help me improve the quality of my academic work."

Table 10/Quality Improvement

Response Value	Meaning	Frequency	Percentage
5	Strongly Agree	16	26.23%
4	Agree	29	47.54%
3	Neutral	14	22.95%
2	Disagree	2	3.28%
1	Strongly Disagree	0	0.00%

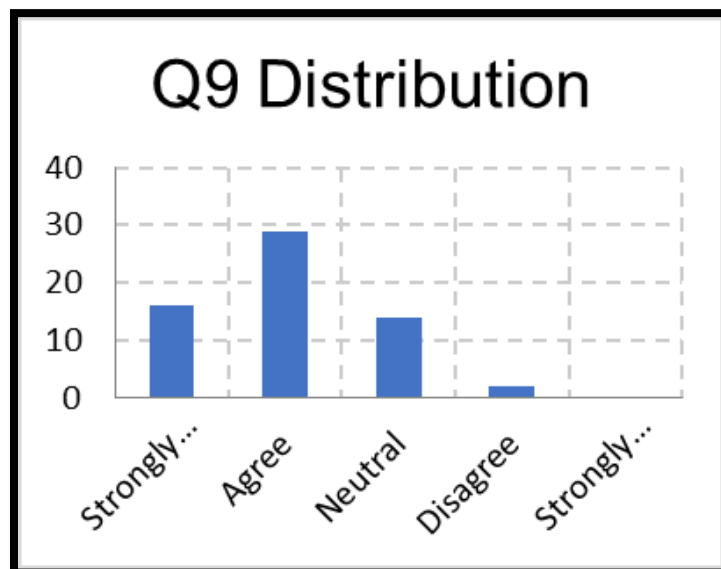


Figure 9/Bar graph for Quality Improvement

Mean Value = 3.97

Interpretation

The mean value of 3.97 indicates agreement. The results show that most respondents believe AI tools help improve the quality of their academic work. This suggests that AI tools support students in producing better academic outputs.

➤ Convenience and Accessibility

Question 10: “AI tools make learning more convenient and accessible.”

Table 11/Convenience and Accessibility

Response Value	Meaning	Frequency	Percentage
5	Strongly Agree	18	29.51%
4	Agree	34	55.74%
3	Neutral	7	11.48%
2	Disagree	2	3.28%
1	Strongly Disagree	0	0.00%

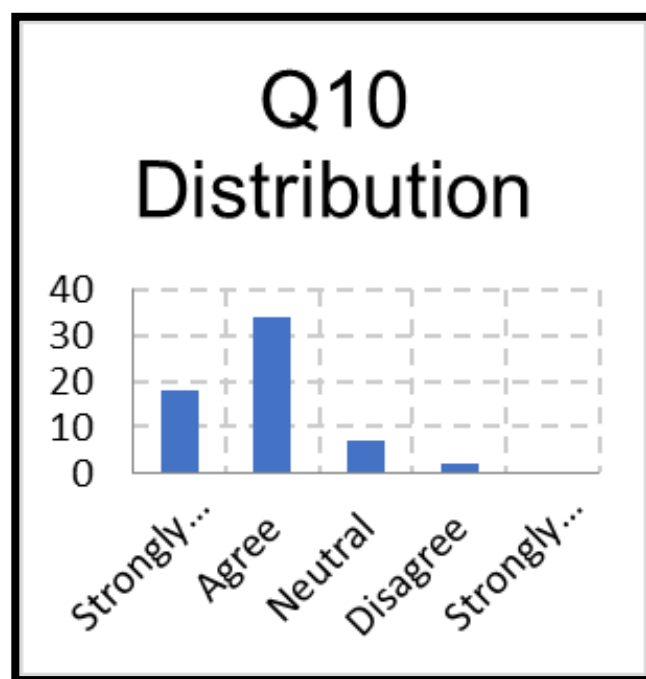


Figure 10/Bar graph for Convenience and Accessibility

Mean Value = 4.11

Interpretation

The mean value of 4.11 indicates strong agreement. Most students agreed that AI tools make learning more convenient and accessible. This shows that AI tools may help students access learning support more easily when needed.

➤ Academic Confidence

Question 11: “I feel more academically confident when using AI tools.”

Table 12/Academic Confidence

Response Value	Meaning	Frequency	Percentage
5	Strongly Agree	15	24.59%
4	Agree	23	37.70%
3	Neutral	17	27.87%
2	Disagree	6	9.84%
1	Strongly Disagree	0	0.00%

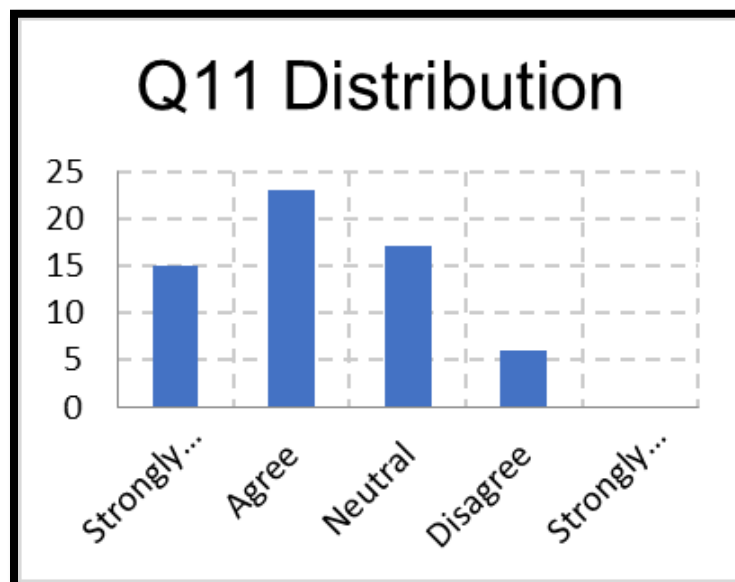


Figure 11/Bar graph for Academic Confidence

Mean Value = 3.77

Interpretation

The mean value of 3.77 indicates agreement. This shows that many students feel more academically confident when using AI tools. However, compared with the other productivity-related questions, this item has a slightly lower mean, suggesting that some students remain neutral or less certain about the effect of AI tools on academic confidence.

➤ Overall Academic Performance

Question 12: “Overall, AI tools have a positive impact on my academic performance.”

Table 13/Overall Academic Performance

Response Value	Meaning	Frequency	Percentage
5	Strongly Agree	14	22.95%
4	Agree	33	54.10%
3	Neutral	12	19.67%
2	Disagree	2	3.28%
1	Strongly Disagree	0	0.00%

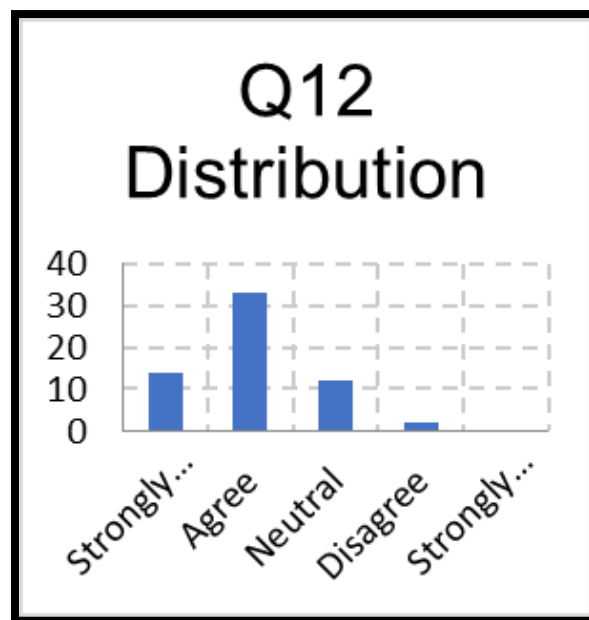


Figure 12/Bar graph for Overall Academic Performance

Mean Value = 3.97

Interpretation

The mean value for this question was 3.97, indicating agreement. The findings show that students generally believe AI tools have a positive impact on their academic performance.

Overall Academic Productivity Mean

Table 14/Overall Academic Productivity Mean

Variable	Questions	Mean	Interpretation
Academic Productivity	Q8–Q12	3.99	Agreement

The overall mean score for academic productivity was 3.99, indicating agreement. This means that respondents generally agreed that AI tools positively contribute to their academic productivity, quality of work, accessibility of learning, academic confidence, and overall academic performance.

Correlation Analysis

Correlation analysis was conducted to examine the relationship between AI tool usage and academic productivity among university students in Sri Lanka. For this analysis, the average score of Q3 to Q7 was used to represent AI tool usage, while the average score of Q8 to Q12 was used to represent academic productivity.

Table 15/Correlation Analysis/Variable

Variable	Mean
AI Tool Usage	4.24
Academic Productivity	3.99

Correlation Result = 0.752

Table 16/Correlation Analysis/Correlation value

Correlation Value	Relationship
0.752	Strong Positive Relationship

Interpretation

The Pearson correlation value was 0.752, which indicates a strong positive relationship between AI tool usage and academic productivity. This means that students who reported higher AI tool usage also tended to report higher academic productivity.

The analysis also showed that the relationship was statistically significant at the 0.05 level, with a p-value of approximately 0.000000000003. Therefore, the result provides strong support for the relationship between AI tool usage and academic productivity among university students in Sri Lanka.

➤ Hypothesis Testing

Table 17/Hypothesis Testing

Hypothesis	Decision
H ₁ : There is a positive relationship between AI tool usage and academic productivity.	Accepted
H ₂ : There is no relationship between AI tool usage and academic productivity.	Rejected

Interpretation

Since the correlation value was positive and strong, the alternative hypothesis H₁ was accepted. Therefore, the findings suggest that there is a positive relationship between AI tool usage and academic productivity among university students in Sri Lanka. The null hypothesis H₀ was rejected.

Summary of Findings

This chapter analyzed the responses collected from 61 university students in Sri Lanka regarding AI tool usage and academic productivity. The demographic analysis showed that the majority of respondents were male, while government and private university students were almost equally represented.

The AI tool usage analysis showed that students strongly agreed with most statements related to using AI tools for academic activities. The overall mean score for AI tool usage was 4.24, indicating strong agreement. The highest mean score was recorded for the statement “I use AI tools to improve my learning efficiency”, with a mean value of 4.36.

The academic productivity analysis showed that students generally agreed that AI tools improve their academic productivity. The overall mean score for academic productivity was 3.99, indicating agreement. The highest productivity-related mean score was recorded for the statement “Using AI tools has improved my academic productivity”, with a mean value of 4.13.

Finally, the correlation analysis showed a strong positive relationship between AI tool usage and academic productivity, with a correlation value of $r = 0.752$. Therefore, the findings confirmed that increased AI tool usage is associated with higher academic productivity among university students in Sri Lanka. As a result, the alternative hypothesis H_1 was accepted, and the null hypothesis H_0 was rejected.

Chapter 5

Discussion

Introduction

This chapter discusses the findings of the study based on the data analysis presented in Chapter 4. The discussion focuses on interpreting the results related to AI tool usage and academic productivity among university students in Sri Lanka. The findings are compared with previous studies discussed in the literature review. In addition, this chapter explains whether the research hypothesis was supported by the findings of the study.

Discussion of Demographic Findings

The demographic analysis revealed that most of the respondents were male students who accounted for 65.57% of the sample while female students made up 31.15% of the respondents. All respondents, except a few who did not want to reveal their gender. The findings also revealed that students of government and private universities were almost equally represented in the study. Such a balanced representation is important as it helps the study to understand the views and experiences of students from different higher education environments in Sri Lanka.

The demographic results show that students from both types of universities are common users of AI tools. The participation of both government and private university students in the study shows that AI technologies are gaining popularity among the Sri Lankan higher education sector. (Creswell, 2014) (Saunders, M., Lewis, P, & Thornhill, A, 2019)

Discussion of AI Tool Usage Findings

The study found that university students often use AI tools for academic activities. The overall mean value for AI tool usage was 4.24, which shows that respondents strongly agreed. The results indicate that students extensively use AI tools in their learning process for academic purposes.

The statement “I use AI tools to improve my learning efficiency” had the highest mean value among the AI usage-related statements with a mean score of 4.36. This suggests that students mainly use AI tools to improve the effectiveness and efficiency of their learning activities. Students also strongly agreed that AI tools help them understand learning materials more effectively and help them manage study time.

These results are in agreement with the previous studies discussed in the literature review. Studies have shown that AI-powered tools such as ChatGPT and Grammarly help students get immediate access to information, simplify complex ideas, and improve their learning experience. Likewise, the current study validates that university students in Sri Lanka consider AI tools as effective educational support systems.

The findings also support the growing trend of using AI in education. As mentioned in the literature review section, today’s university students are more and more reliant on digital technologies for academic support. The findings of this study also support the fact that AI tools have become embedded in the daily academic activities of students. (Kasneci, E, et al., 2023) (UNESCO, 2021) (Popenici, S. A. D & Kerr, S., 2017)

Discussion of Academic Productivity Findings

The study’s findings indicate that AI tools have a positive impact on the academic productivity of university students. The overall mean score of academic productivity is 3.99 which indicates agreement of the respondents. This shows that students perceive AI tools as contributing positively to their academic performance and productivity in general.

The statement “Using AI tools has improved my academic productivity” had the highest mean value of 4.13. This suggests that students think that AI tools assist them in doing academic work better and faster. Students also agreed that AI tools improve the quality of academic work, increase learning accessibility and enhance academic confidence.

However, the statement related to academic confidence recorded a slightly lower mean value of 3.77 compared to other productivity-related statements. This implies that while a large number of students find AI tools helpful for learning, there may still be some students who are not sure about relying on AI technologies for academic use.

These results are in line with the previous studies that were reviewed in the literature review. Previous studies indicated that AI tools can increase learning efficiency, save time and help students to complete academic tasks more efficiently. The current study extends these findings in a sample of Sri Lankan university students.

At the same time, the study's results also echo concerns raised in the literature review regarding over-reliance on AI tools. While students generally had positive experiences, some neutral responses suggest that not all students trust or rely on AI systems for learning activities. (Luckin, R, Holmes, W, Griffiths, M, & Forcier, L. B, 2016) (Selwyn, 2019) (Zawacki-Richter, O, Marín, V. I, Bond, M., & Gouverneur, F., 2019)

Discussion of Correlation Analysis

One of the major purposes of conducting this research was to assess the relation between the use of AI tools and academic productivity. Results from the Pearson correlation analysis indicate that there is a highly positive correlation between these variables and its value is equal to:

$$r = 0.752$$

This means that students who used more AI tools were also found to be more academically productive. Thus, the research findings support the hypothesis stating that there is a connection between the use of AI tools and increased productivity of university students in Sri Lanka.

These findings are important in relation to the purpose of the paper since they clearly show that AI tools positively influence learning efficiency and academic activity.

As indicated earlier, previous literature on this topic found positive connections between technological tools and academic performance. The reasons were seen in students' ability to gain fast access to information, learn conveniently and perform educational tasks more efficiently. Similarly, the current research shows that AI tools may contribute to productivity. (Dwivedi, Y. K., et al., 2023) (Tlili, A, et al., 2023)

Hypothesis Discussion

For testing purposes, the following hypothesis was used in the experiment:

- H_1 : There is a positive relationship between AI tool usage and academic productivity.
- H_0 : There is no relationship between AI tool usage and academic productivity.

As a result of correlation analysis where the correlation coefficient of 0.752 was obtained, there exists a significant positive relationship between AI tool usage and academic productivity. Accordingly, the alternative hypothesis H_1 is accepted whereas the null hypothesis H_0 is rejected.

By accepting the alternative hypothesis (H_1), it can be concluded that there is a positive relationship between AI tool usage and students' academic productivity. It implies that students who make use of AI tools have higher chances of having learning productivity. (Creswell, 2014) (Sekaran, U & Bougie, R, 2016)

Summary

In this chapter, the research results concerning the impact of AI tools on academic productivity in Sri Lankan university students were presented. According to the results, university students often use AI tools in their academic activities and believe that using these tools can enhance their learning efficiency, academic productivity, and availability of learning services.

Also, according to the results, there is a positive correlation between the use of AI tools and academic productivity of university students. This means that the increased use of such tools leads to higher academic productivity. Moreover, the results are in line with the information presented in the literature review.

Summarizing, one can state that AI tools are widely used by modern university students and play an important role in their academic activities and productivity. (Kasneci, E, et al., 2023) (UNESCO, 2021)

Chapter 6

Conclusion and Recommendations

Conclusion

This study investigated the impact of AI tools on the academic productivity of university students in Sri Lanka. The main aim of the research was to determine whether the application of AI tools has a positive impact on students' academic productivity. To achieve this objective, data were collected from 61 university students through questionnaire and analyzed using Microsoft Excel.

The results of the study showed that university students used AI tools for academic activities. Most respondents strongly agreed that AI tools help improve learning efficiency, support understanding of learning materials, help manage study time, and help complete academic tasks more quickly. The average overall score for AI tool usage was 4.24, reflecting strong agreement among respondents.

The findings also indicated that AI tools have a positive impact on the academic productivity. Mostly, the students agreed that the AI tools increased the convenience of learning, the quality of the academic work, the academic productivity and the positive impact on the academic performance. The means of the general academic productivity were 3.99 which were in agreement among the respondents.

Furthermore, correlation analysis revealed a strong positive correlation between the usage of AI tool and academic productivity with a Pearson correlation value of:

$$r = 0.752$$

The finding confirmed that higher academic productivity among university students in Sri Lanka was associated with increased use of AI tools. Therefore, the null hypothesis (H0) was rejected and the alternative hypothesis (H1) was accepted.

Overall, the research highlights the importance of AI tools in assisting students' academic activities and productivity in modern higher education environments. Students should, however, use AI technologies responsibly to ensure that learning remains meaningful and effective.

Recommendations

Based on the results of this study, several recommendations are given to students, universities, lecturers, and future educational practices related to the implementation of AI tools in higher education. The recommendations are intended to maximize the academic benefits of AI technologies and minimize the risks of overreliance and unethical use.

First, students should employ AI tools ethically, as supportive learning assistants and not as complete substitutes for independent learning. AI tools may boost productivity, but students still need to build their own analytical thinking, creativity, problem-solving skills and subject knowledge. Over-reliance on AI-generated content could be detrimental to the critical thinking and long-term learning development of students. Students should use AI technologies to help with learning, but still be active participants in academic activities.

Second, universities must develop clear policies and ethical guidelines for the academic use of AI tools. As AI technologies are increasingly used in higher education, universities need to establish proper rules to make sure that their students use them in a responsible and fair way. Clear institutional guidelines can foster productive and ethical learning practices while helping to prevent academic misconduct, plagiarism and misuse of AI-generated content.

Third, universities and lecturers should hold workshops, seminars and awareness programs to educate students on the benefits, limitations and risks of AI tools in education. A lot of students use AI tools frequently, but some may not fully grasp the ethical implications and academic risks associated with relying too heavily on AI technologies. Awareness programs can help students learn to use AI tools in a way that maintains academic integrity and independent learning skills.

Moreover, lecturers should seek effective ways to integrate AI technologies into teaching and learning activities. Rather than banning AI tools altogether, educators can teach students to use the technologies in a productive way to support their learning, generate ideas, support learning and develop skills. The successful integration of AI technologies can raise the quality of education, the accessibility of learning resources and student engagement in higher education.

Moreover, students should strike a balance between AI use and self-learning practices. AI tools may be time-saving and convenient, but students should still do their own reading, research, writing and critical analysis. This balance is important to keep in order to develop strong academic and professional competencies for future careers for students.

At last, universities need to foster a culture of responsible AI within higher education institutions. Educational institutions can promote ethical AI practices through awareness campaigns, academic assistance programs, and digital literacy initiatives related to AI technologies. The use of responsible AI can help students enjoy the benefits of technological progress while minimizing potential risks such as cheating, misinformation, and over-reliance on AI systems.

The following table summarizes the key recommendations developed based on the findings of the study.

Recommendations Based on Study Findings

Table 18/Recommendations Based on Study Findings

Identified Issue / Finding	Detailed Recommendation
Students highly depend on AI tools for academic activities	Universities should educate students on responsible AI usage and encourage them to use AI tools as learning support systems rather than complete replacements for independent thinking and academic work.
AI tools improve learning efficiency and productivity	Educational institutions should integrate AI-supported learning methods into academic environments while ensuring proper monitoring and ethical usage practices.
Some students are uncertain about academic confidence when using AI tools	Lecturers should guide students on balancing AI usage with self-learning, critical thinking, and independent academic skill development.
Increasing use of AI tools in higher education	Universities should develop formal policies, ethical standards, and academic guidelines related to AI usage in education.
Students use AI tools for faster task completion	Students should be encouraged to maintain creativity, problem-solving skills, and active learning practices while using AI technologies.
Possibility of overdependence on AI-generated content	Universities should conduct awareness programs and workshops to educate students about the risks of excessive AI dependence and academic dishonesty.
Lack of awareness regarding ethical AI usage	Educational institutions should introduce AI literacy programs to improve students' understanding of ethical and responsible AI practices.
AI tools are becoming common among university students	Universities should continuously evaluate the impact of AI technologies on teaching, learning, and student performance to ensure positive educational outcomes.

Future Research

This study was conducted in Sri Lanka and the sample was 61 respondents from university students. Expanding the study with larger sample sizes will allow future researchers to enhance the generalizability and reliability of the findings.

Additionally, future research can compare the use of AI tools and academic productivity among students from different countries, universities, or academic disciplines. These comparative studies can shed more comprehensive insights into the impact of AI technologies on higher education across different contexts.

In addition, future research can also explore other factors related to the use of AI in education such as students' satisfaction, learning outcomes, academic performance, critical thinking skills, and ethical issues of AI technologies.

Researchers can also use advanced statistical tools and more complex research methods to further examine the relationship between AI tool usage and academic productivity. Qualitative studies involving interviews and focus group discussions may also give deeper understanding of students' experiences and attitudes toward AI tools in education.

Overall, future research can continue exploring the growing role of Artificial Intelligence in higher education and its long-term impact on students and learning environments.

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Appendices

Appendix A

Google Form Questionnaire

The screenshot shows a Google Form interface. At the top, there are tabs for 'Questions', 'Responses' (with a count of 61), and 'Settings'. A yellow banner across the top states 'This form isn't accepting responses.' with a 'Manage' link. The form title is 'The Impact of AI Tools on Academic Productivity Among University Students in Sri Lanka'. Below the title is a text box with a rich text editor toolbar (B, I, U, link, unlink) and the text: 'This questionnaire is conducted to examine the impact of AI tools on academic productivity among university students in Sri Lanka. Your responses will be kept confidential and used only for academic research purposes.' Below this is a section titled 'Demographic Information' with the text: 'This section collects general background information about the participants for research purposes only. All responses will remain confidential.' The first question is '1. What type of university are you currently studying in? *' with two radio button options: 'Government University' and 'Private University'. The second question is '2. What is your gender? *' with two radio button options: 'Male' and 'Female'. On the right side of the form, there is a vertical toolbar with icons for adding questions, duplicating, deleting, inserting a table, inserting a video, and inserting a grid.

Figure 13/Google Form Questionnaire

Appendix B

Sample Dataset

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
1	Timestamp	1. What ty	2. What i	3. I frequ	4. AI tools	5. I use A	6. AI tool	7. AI tool	8. Using A	9. AI tool	10. AI toc	11. I feel	12. Overall, AI tools have a positive impact on my academic performance.						
2	2026/05/1	Private Ur	Male	Agree	Strongly A	Agree	Agree	Agree	Strongly A	Agree	Neutral	Neutral	Agree						
3	2026/05/1	Private Ur	Male	Agree	Agree	Agree	Agree	Agree	Neutral	Neutral	Strongly A	Agree	Agree						
4	2026/05/1	Private Ur	Prefer not	Neutral	Agree	Neutral	Agree	Neutral	Neutral	Neutral	Disagree	Disagree	Neutral						
5	2026/05/1	Private Ur	Male	Agree	Agree	Agree	Agree	Agree	Agree	Agree	Agree	Agree	Agree						
6	2026/05/1	Governm	Male	Strongly A	Strongly A	Agree	Strongly A	Agree	Agree	Agree	Neutral	Neutral	Neutral						
7	2026/05/1	Private Ur	Male	Strongly A	Agree	Strongly A	Strongly A	Strongly A	Agree	Agree	Strongly A	Agree	Agree						
8	2026/05/1	Governm	Male	Agree	Strongly A	Strongly A	Strongly A	Strongly A	Strongly A	Strongly A	Strongly A	Strongly A	Agree						
9	2026/05/1	Governm	Female	Strongly A	Strongly A	Strongly A	Strongly A	Strongly A	Strongly A	Strongly A	Strongly A	Agree	Agree	Agree					
10	2026/05/1	Private Ur	Female	Strongly A	Strongly A	Strongly A	Strongly A	Strongly A	Strongly A	Strongly A	Strongly A	Strongly A	Strongly A	Strongly A	Agree				
11	2026/05/1	Governm	Male	Strongly A	Strongly A	Strongly A	Strongly A	Strongly A	Strongly A	Strongly A	Strongly A	Strongly A	Strongly A	Strongly A	Strongly A	Agree			
12	2026/05/1	Private Ur	Female	Agree	Agree	Agree	Disagree	Agree	Agree	Agree	Agree	Neutral	Agree						
13	2026/05/1	Private Ur	Female	Agree	Agree	Agree	Agree	Neutral	Agree	Strongly A	Neutral	Neutral	Neutral						
14	2026/05/1	Governm	Prefer not	Agree	Agree	Agree	Agree	Agree	Agree	Neutral	Agree	Agree	Agree						
15	2026/05/1	Governm	Male	Neutral	Neutral	Disagree	Disagree	Neutral	Neutral	Disagree	Agree	Neutral	Neutral						
16	2026/05/1	Governm	Female	Strongly A	Agree	Neutral	Agree	Agree	Agree	Agree	Agree	Neutral	Neutral						
17	2026/05/1	Private Ur	Male	Strongly A	Strongly A	Strongly A	Strongly A	Strongly A	Strongly A	Strongly A	Strongly A	Agree	Agree	Agree	Agree				
18	2026/05/1	Private Ur	Male	Disagree	Neutral	Neutral	Neutral	Neutral	Agree	Neutral	Neutral	Disagree	Neutral						
19	2026/05/1	Private Ur	Female	Agree	Strongly A	Agree	Strongly A	Agree	Agree	Disagree	Agree	Agree	Agree						
20	2026/05/1	Governm	Male	Strongly A	Strongly A	Strongly A	Strongly A	Strongly A	Strongly A	Strongly A	Strongly A	Strongly A	Strongly A	Strongly A	Agree				
21	2026/05/1	Private Ur	Male	Agree	Strongly A	Strongly A	Agree	Strongly A	Disagree	Agree	Agree	Agree	Strongly A	Agree					
22	2026/05/1	Governm	Male	Agree	Neutral	Strongly A	Neutral	Disagree	Agree	Neutral	Agree	Disagree	Neutral						
23	2026/05/1	Governm	Male	Strongly A	Strongly A	Agree	Strongly A	Strongly A	Agree	Strongly A	Agree	Neutral	Agree						
24	2026/05/1	Private Ur	Female	Agree	Strongly A	Strongly A	Agree	Strongly A	Agree	Agree	Agree	Neutral	Agree						
25	2026/05/1	Governm	Female	Strongly A	Agree	Strongly A	Agree	Strongly A	Strongly A	Agree	Agree	Strongly A	Strongly A	Agree	Strongly A	Strongly A	Agree		
26	2026/05/1	Private Ur	Female	Agree	Neutral	Agree	Agree	Neutral	Neutral	Neutral	Agree	Disagree	Agree						
27	2026/05/1	Private Ur	Male	Neutral	Agree	Agree	Agree	Neutral	Agree	Agree	Agree	Agree	Agree						
28	2026/05/1	Governm	Female	Agree	Agree	Agree	Agree	Agree	Agree	Agree	Agree	Agree	Agree						
29	2026/05/1	Private Ur	Female	Agree	Agree	Agree	Neutral	Neutral	Agree	Neutral	Agree	Neutral	Neutral						
30	2026/05/1	Governm	Female	Strongly A	Strongly A	Strongly A	Strongly A	Agree	Agree	Strongly A	Strongly A	Disagree	Neutral						
31	2026/05/1	Private Ur	Female	Agree	Agree	Agree	Agree	Agree	Agree	Agree	Agree	Agree	Agree						
32	2026/05/1	Private Ur	Male	Agree	Agree	Strongly A	Strongly A	Strongly A	Agree	Neutral	Agree	Agree	Agree						
33	2026/05/1	Private Ur	Female	Strongly A	Strongly A	Strongly A	Strongly A	Strongly A	Strongly A	Strongly A	Strongly A	Strongly A	Strongly A	Strongly A	Strongly A	Strongly A	Strongly A	Agree	
34	2026/05/1	Private Ur	Female	Strongly A	Strongly A	Strongly A	Strongly A	Strongly A	Strongly A	Strongly A	Strongly A	Strongly A	Strongly A	Strongly A	Agree				
35	2026/05/1	Private Ur	Male	Strongly A	Agree	Strongly A	Agree	Agree	Agree	Agree	Agree	Neutral	Agree						
36	2026/05/1	Private Ur	Male	Strongly A	Strongly A	Strongly A	Strongly A	Strongly A	Strongly A	Strongly A	Strongly A	Strongly A	Strongly A	Strongly A	Agree				
37	2026/05/1	Governm	Male	Agree	Agree	Strongly A	Agree	Agree	Strongly A	Agree	Strongly A	Disagree	Disagree						
38	2026/05/1	Governm	Male	Strongly A	Strongly A	Strongly A	Strongly A	Strongly A	Strongly A	Strongly A	Strongly A	Strongly A	Strongly A	Strongly A	Strongly A	Strongly A	Agree		

Figure 14/Sample Dataset

Appendix C

Correlation Analysis

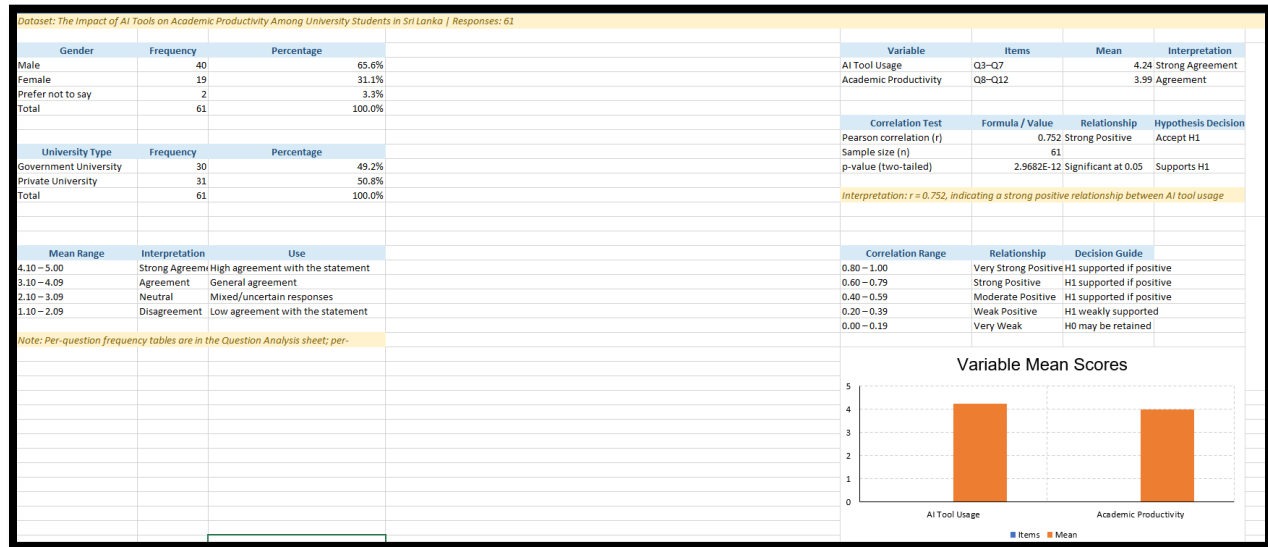


Figure 15/Correlation Analysis